



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Midterm Exam, Trimester: Summer 2024

Course Code: CSE-1115, Course Title: Object Oriented Programming

Total Marks: 30, Duration: 1 Hour 45 Minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all Five Questions

QUESTION 1

[2+2+2 MARKS]

Consider the following codes:

```
public class Person {  
    public String name, gender;  
    private int age; // private = restricted access  
  
    // Write constructor to initialize name and gender with this reference keyword;  
  
    // Write getter method for age;  
  
    // Write setter method for age;  
  
    public static void main(String[] args) {  
        Person p1 = new Person("Prof. Albert Einstein", "Male");  
        System.out.println(p1.name);  
    }  
}
```

Now:

- I. Write constructor to initialize name and gender with this reference keyword.
- II. Write getter method for age variable.
- III. Write setter method for age variable.

QUESTION 2

[3+3 MARKS]

Consider the following codes:

```
class Vehicle {  
    protected String brand;  
  
    public Vehicle(String brand) {  
        this.brand = brand;  
    }  
  
    public void honk() {  
        System.out.println("Tuut, tuut!");  
    }  
}
```

Output:
A vehicle horn is a sound-making device.
Tuut, tuut!
Ford Mustang

<pre> public class Car extends Vehicle { private String modelName; // Invoke parent class constructor; // Invoking overriding method; public static void main(String[] args) { Car myCar = new Car("Ford", "Mustang"); myCar.honk(); System.out.println(myCar.brand + " " + myCar.modelName); } } </pre>	
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Now:

- I. Write the constructor of car class and invoke the parent class constructor.
- II. Override the honk() method and print “A vehicle horn is a sound-making device.” then invoke the overridden method.

QUESTION 3

[6 MARKS]

Write output of the following codes:

```

class Calculate{
    static int count=10;

    static{ System.out.println("United International University"); }

    static int cube(int x){
        return x*x*x;
    }

    public static void Counter(){
        count++;//incrementing the value of static variable
        System.out.println(count);
    }

    public static void main(String args[]){
        Calculate.Counter();
        Calculate c1 = new Calculate();
        c1.Counter();
        Calculate c2 = new Calculate();
        c2.Counter();
        System.out.println(c1.count);
        int result=Calculate.cube(Calculate.count);
        System.out.println(result);
    }
}

```

QUESTION 4

[6 MARKS]

Write output of the following codes:

```
public class Sum {
    int x, y, z;

    {
        System.out.println(x+y+z);
    }

    {
        x = 10;
        y = 20;
        z = 30;
        System.out.println(x+y+z);
    }

    public Sum(int x) {
        this.x = x;
        System.out.println(x+y+z);
    }

    public Sum(int x, int y) {
        this(1000);
        this.x = x;
        this.y = y;
        System.out.println(x+y+z);
    }

    public Sum(int x, int y, int z) {
        this(100, 200);
        this.x = x;
        this.y = y;
        this.z = z;
        System.out.println(x+y+z);
    }

    public void display(){
        this.display("That's one small step for man, one giant leap for mankind.");
    }

    public void display(String str){
        System.out.println(str);
    }

    public static void main(String args[])
    {
        Sum s = new Sum(10, 20, 30);
        s.display();
    }
}
```

QUESTION 5**[6 MARKS]**

Consider the following Java code:

<pre>class Shape { void draw() { System.out.println("drawing..."); } } class Rectangle extends Shape { void draw() { System.out.println("drawing rectangle..."); } } class Circle extends Shape { void draw() { System.out.println("drawing circle..."); } } class Triangle extends Shape { void draw() { System.out.println("drawing triangle..."); } }</pre>	Output: drawing rectangle... drawing circle... drawing triangle...
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Write a class named Test, which has the main() method. In the main() method, create a heterogeneous array named "s" of Shape class, where array length is three. Index s[0] creates with Rectangle class, s[1] creates with Circle class, s[2] creates with Triangle class. Finally, iterate a loop to call draw() method of each object.